INCH-POUND

MIL-C-915/85A
29 September 1989
SUPERSEDING
MIL-C-915/85
18 June 1985

MILITARY SPECIFICATION SHEET

CABLE, POWER ELECTRICAL, THREE TWISTED PAIRS NO. 16 GAUGE, FLUOROPOLYMER INSULATION, FOR SUBMARINE OUTBOARD USE ONLY (NOT FOR INBOARD USE), TYPE 3PR-16

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specificed in the solicitation: MIL-C-915.

REQUIREMENTS:

Qualification required.

Construction

First - Copper conductor, tin coated, AWG size 16, class B stranding.
Second - Insulation of ethylene tetrafluoroethylene (ETFE). Nominal wall thickness 0.010 inch. Maximum overall diameter 0.082 inch.
Third - Two insulated conductors, one black and one white, twisted together with a lay of 1-1/4 ± 1/8 inch.
Fourth - Marker braid on each pair, standard identification code applied by method 4.
Fifth - Three pairs cabled with a lay not greater than 24 times the pitch diameter of the layer.
Sixth - Rubber filler in order to form a firm, round assembly. Filler material shall be capable of removal from the assembly without damaging the insulation on conductors.
Seventh - Polyester binder tape applied helically with overlap.

AMSC N/A FSC 6145
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REQUIREMENTS - Continued

Eighth - Cable jacket of black polyether polyurethane. The cable jacket material shall be B.F. Goodrich Co. ESTANE 58300 or 58863; or Mobay Chemical Co. TEXIN 985A. Nominal wall thickness 0.100 inch. Overall cable diameter to be 0.530 to 0.550 inch.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - at 20°C, ohms/1000 feet, maximum	4.24
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2000
Conductor to water	2000
Insulation dielectric (on 100 percent of insulated conductor) - test in accordance with MIL-W-81381, using a test voltage of 8.0 kV (peak), minimum.	
Insulation resistance - megohms/1000 feet, minimum	5000
Group A:	
Visual and dimensional	No failure
Group B:	
Cold bending, cable - at minus 40 ± 2°C over a 3-inch diameter mandrel	No damage
Physicals (unaged)	
Insulation	
Tensile strength - lbf/in², minimum	5000
Elongation - percent, minimum	100
Cable jacket	
Tensile strength - lbf/in², minimum	4000
Elongation - percent, minimum	500
Group C:	
Physicals (aged)	
Insulation	
Air oven (168 hours at 180°C)	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged, minimum.	90
Cable jacket	
Air oven (168 hours at 80°C)	
This test shall be in accordance with MIL-C-915	No failure

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EXAMINATION AND TESTS - Continued

	Requirements
Hot oil immersion	
Tensile strength, period of unaged, minimum	60
Elongation, percent of unaged, minimum	80

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, plus all of groups A, B and C.

UNIT ORDERING LENGTH: 500 feet (minimum).

NOTE: Not for Air Force use.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - MI
Navy - SH

Preparing activity:
Navy - SH
(Project 6145-1127-22)

Review activities:
Army - CR, ER
Navy - EC
DLA - IS

User activities:
Army - AL, AR, ME
Navy - CG

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